WORKED SOLUTION

STEP I 2000, Question 9

Mechanics • Projectile motion

STEP I 2000, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s1-q9

The key idea

The range is measured *in the carriage*, so work in the carriage's frame. Since the carriage accelerates north at a, the shell's acceleration relative to the floor is g downwards plus a southwards, which is the direction the cannon fires in. The motion is still a projectile with constant acceleration, just with an extra horizontal term. Once the range is written in terms of 2θ, maximising it is a one-line differentiation.

Solution

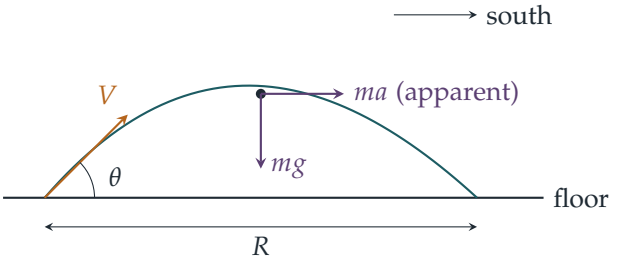
The range in the carriage

Model the shell as a particle and ignore air resistance. Assume it is fired from floor level and lands on the floor, and that the carriage is long enough for it to land inside. Take x horizontal and pointing south (along the firing direction) and y vertically up, both measured relative to the carriage.

In the ground frame the shell has acceleration g downwards and no horizontal acceleration. The carriage has acceleration a northwards. So the shell's acceleration *relative to the carriage* is

$$\ddot{x} = 0 - (-a) = a, \qquad \ddot{y} = -g.$$

Equivalently, in the carriage's frame there is an apparent force ma southwards on a shell of mass m, as well as its weight.



The initial velocity relative to the carriage is $V\cos\theta$ south and $V\sin\theta$ up, so

$$x = Vt\cos\theta + \tfrac{1}{2}at^2, \qquad y = Vt\sin\theta - \tfrac{1}{2}gt^2.$$

The vertical motion is unchanged by a: the shell lands when $y = 0$ with $t > 0$, that is at $T = \dfrac{2V\sin\theta}{g}$. Its range in the carriage is therefore

$$\begin{aligned} R = x(T) &= \frac{2V^2\sin\theta\cos\theta}{g} + \frac{a}{2}\cdot\frac{4V^2\sin^2\theta}{g^2} \\ &= \frac{V^2}{g}\left(\sin 2\theta + \frac{a}{g}(1-\cos 2\theta)\right), \end{aligned}$$

using $2\sin\theta\cos\theta = \sin 2\theta$ and $2\sin^2\theta = 1 - \cos 2\theta$.

Maximising the range

Differentiating with respect to θ,

$$\frac{\mathrm{d}R}{\mathrm{d}\theta} = \frac{2V^2}{g}\left(\cos 2\theta + \frac{a}{g}\sin 2\theta\right),$$

which is zero when

$$\boxed{\tan 2\theta = -\frac{g}{a}.}$$

? **Ask yourself** A negative tangent? Does that mean something has gone wrong?

No. For $0 < \theta < \frac{\pi}{2}$ we have $0 < 2\theta < \pi$, and $\tan 2\theta < 0$ simply puts 2θ in $(\frac{\pi}{2}, \pi)$, so $\theta > \frac{\pi}{4}$. That makes sense: the extra push a acts for as long as the shell is in the air, so it pays to fire a little higher than 45° and buy more flight time. With $a = 0$ we would have $\cos 2\theta = 0$, giving the familiar $\theta = \frac{\pi}{4}$.

To see that this is a maximum, write $\tan\varphi = \dfrac{a}{g}$ with $0 < \varphi < \frac{\pi}{2}$. Then

$$\sin 2\theta - \frac{a}{g}\cos 2\theta = \frac{\sin(2\theta - \varphi)}{\cos\varphi}, \qquad \text{so} \qquad R = \frac{V^2}{g}\left(\frac{\sin(2\theta-\varphi)}{\cos\varphi} + \frac{a}{g}\right).$$

This is largest when $2\theta - \varphi = \frac{\pi}{2}$, that is when $2\theta = \frac{\pi}{2} + \varphi$, which is the root of $\tan 2\theta = -g/a$ in $(\frac{\pi}{2}, \pi)$. So the optimal angle is exactly

$$\theta = \frac{\pi}{4} + \frac{1}{2}\arctan\frac{a}{g},$$

and, since $\cos\varphi = g/\sqrt{g^2 + a^2}$, the maximum range is exactly

$$R_{\max} = \frac{V^2}{g}\left(\frac{\sqrt{g^2+a^2}}{g} + \frac{a}{g}\right) = \frac{V^2}{g^2}\left(a + \sqrt{g^2+a^2}\right).$$

Small a

For small u, $\arctan u = u + O(u^3)$. With $u = a/g$,

$$\theta = \frac{\pi}{4} + \frac{1}{2}\left(\frac{a}{g} + O\left(\frac{a^3}{g^3}\right)\right) \approx \frac{\pi}{4} + \frac{a}{2g},$$

as required. For the range, $\sqrt{g^2 + a^2} = g\sqrt{1 + a^2/g^2} = g + O(a^2/g)$, so

$$R_{\max} = \frac{V^2}{g^2}(a + g + O(a^2/g)) \approx \frac{V^2}{g} + \frac{V^2 a}{g^2},$$

which is the second required result. Both approximations are correct to first order in a/g; the neglected terms are of order $(a/g)^2$ or smaller.

If you prefer to stay with $\tan 2\theta$, write $2\theta = \frac{\pi}{2} + \varepsilon$. Then $\tan 2\theta = -\cot\varepsilon \approx -1/\varepsilon$, so $\varepsilon \approx a/g$. Substituting $\sin 2\theta = \cos\varepsilon \approx 1$ and $\cos 2\theta = -\sin\varepsilon \approx -\varepsilon$ into R gives $R \approx \frac{V^2}{g}(1 + \frac{a}{g})$ to first order.

Checks

- $a = 0$: the formulae reduce to $\theta = \frac{\pi}{4}$ and $R_{\max} = V^2/g$, the standard results on level ground.
- Dimensions: a/g is dimensionless and V^2/g is a length, as it should be.
- Numerically, with $V = 7$, $g = 9.8$ and $a = 1.3$, stepping the motion in the ground frame and subtracting the carriage's position reproduces the formula for R (e.g. $R \approx 5.478$ at $\theta = 0.7$). A brute-force search over θ for several random a found the maximum where $\tan 2\theta = -g/a$, with $R_{\max}$ agreeing with $\frac{V^2}{g^2}(a + \sqrt{g^2 + a^2})$ to 10 significant figures. For $a = 0.05$ the exact optimal angle $0.787949\ldots$ matches $\frac{\pi}{4} + \frac{a}{2g}$ to seven decimal places.

Common slips

- Getting the sign of a wrong. The carriage accelerates *away* from the shell's direction of fire, so relative to the floor the shell gains speed southwards: the term $\frac{1}{2}aT^2$ is added, not subtracted.
- Measuring the range on the ground rather than in the carriage. The question asks for the distance along the carriage floor.
- Taking $2\theta = \arctan(-g/a)$ from a calculator, which gives a negative angle. The correct root lies in $(\frac{\pi}{2}, \pi)$.

Beyond STEP

Working in an accelerating frame and adding an apparent force $-m\mathbf{a}$ is the simplest example of a *fictitious force*. Rotating frames bring the centrifugal and Coriolis forces

in the same way. Here the apparent force simply tilts "effective gravity" to magnitude $\sqrt{g^2+a^2}$, which is one way to see why the answer contains $\sqrt{g^2+a^2}$. Non-inertial frames are treated clearly in Kleppner and Kolenkow, *An Introduction to Mechanics*.

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